

HEADACHE

Give urgent attention to the patient with headache and any of:

- Decreased consciousness →16
- BP ≥ 180/110 and not pregnant →132
- Pregnant or 1 week postpartum, and BP ≥ 140/90 →159
- Sudden weakness/numbness of face/arm/leg or speech problem →136
- New vision problems or eye pain →31

- Sudden severe headache or dizziness
- Headache that is getting worse and more frequent
- Headache that wakes patient or is worse in the morning
- Neck stiffness, drowsy/confused or purple/red rash: **meningitis** likely
- Persistent nausea/vomiting

- Persistent headache since starting ART
- Following a first seizure
- Recent head injury
- Unequal pupils

Manage and refer urgently:

- If temperature ≥ 38°C or **meningitis** likely: give **ceftriaxone** 2g IV¹/IM. Avoid injecting > 1g IM at one injection site.
- If in a malaria area in past 3 months and malaria test² positive: give **artesunate** 2.4mg/kg IM. If artesunate unavailable, give quinine as slow IV infusion over 4 hours: dilute quinine 20mg/kg in 5% dextrose 5-10mL/kg. If IV not possible, give IM³ diluted in **sodium chloride 0.9%**.

Approach to the patient with headache not needing urgent attention

Does patient have fever and body pain or recent common cold?

Yes

Has patient had recent common cold and now any of: thick nasal/postnasal discharge, pain when pushing on forehead/cheeks, headache worse on bending forward?

Yes

Sinusitis likely

- Give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) for up to 5 days.
- Give **sodium chloride 0.9%** nose drops as needed.
- Give **oxymetazoline 0.05%** 2 drops in each nostril 8 hourly for up to 5 days. Advise against overuse which may worsen blocked nose.
- If symptoms ≥ 10 days, fever ≥ 38°C, purulent nasal discharge, facial pain ≥ 3 days, or symptoms worsen after initial improvement, give **amoxicillin** 500mg 8 hourly for 5 days. If severe penicillin allergy⁴, give instead **azithromycin** 500mg daily for 3 days.
- If recurrent, test for HIV →110.
- If tooth infection or swelling over sinus/around eye, refer same day.

No

- If in a malaria area in past 3 months, arrange same day malaria test². If positive, **malaria** likely →24.
- If patient has a tick bite (small dark brown/black scab) or tick present, **tick bite fever** likely →24.
- If none of above, treat as **acute viral infection**:
 - Consider COVID-19 →40. Advise patient to isolate at home for 7 days from start of his/her symptoms.
 - Advise to wear mask indoors and social distance. While unwell, avoid contact with elderly/those with chronic diseases/groups, wash hands.
 - Give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) as needed for up to 5 days. Explain antibiotics are not needed. Advise to rest and maintain hydration.
 - Advise to return if worsening symptoms: if cough/difficulty breathing →38, if face pain →32, if ear pain →33.

No

Does patient get recurrent headaches that are throbbing, disabling with nausea or light/noise sensitivity, that resolve completely within 72 hours?

Yes

Migraine likely

- Give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) or **ibuprofen**⁵ 400mg 8 hourly with food for up to 5 days.
- If nausea, also give **metoclopramide** 10mg 8 hourly up to 3 doses.
- Advise to recognise and treat migraine early, rest in dark, quiet room.
- Advise regular meals, keep hydrated, regular exercise, good sleep hygiene.
- Keep a headache diary to identify triggers like lack of sleep, hunger, stress, caffeine, chocolate, cheese. Avoid if possible.
- Avoid oestrogen-containing contraceptives →154.
- If ≥ 2 attacks/month, refer/discuss for medication to prevent migraines.

No

- Check BP. If ≥ 140/90 →132.
- Ask about type and site of pain:

Tightness around head or generalised pressure-like pain

Tension headache likely

- Give **paracetamol** 1g 6 hourly as needed for up to 5 days.
- Assess for stress and anxiety →86.
- Advise regular exercise.

Constant aching pain, tender neck muscles

Muscular neck pain likely →64.

Patient > 50 years, pain over temples

Giant cell arteritis likely

- Check CRP.
- Give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) for up to 5 days.
- Review next day: if CRP > 5, discuss with specialist same day.

Advise to use analgesia only when necessary. Overuse may cause headaches: if using analgesia > 2 days/week for ≥ 3 months, advise to reduce amount used. Headache should improve within 2 months of decreased use.

If diagnosis uncertain or poor response to treatment, discuss/refer.

¹Do not mix Ringer's lactate and IV ceftriaxone. Flush IV line with **sodium chloride 0.9%** before and after IV ceftriaxone. ²Test for malaria with rapid diagnostic test if available, and parasite slide microscopy. ³To give IM quinine: first calculate volume of sodium chloride 0.9% in mL: weight × 20 ÷ 100. Then add this volume of sodium chloride 0.9% to quinine 20mg/kg and inject half the volume into each thigh. ⁴History of anaphylaxis, urticaria or angioedema. ⁵Avoid ibuprofen if peptic ulcer, asthma, hypertension, heart failure, kidney disease.